

SONORA, TX, USA

WMO: 747335

Lat: 30.586N

Lon: 100.649W

Elev: 652

StdP: 93.73

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03059

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.9	(c) -3.0	(d) -13.7	(e) 1.2	(f) 2.9	(g) -12.1	(h) 1.4	(i) 4.6	(j) 9.9	(k) 8.8	(l) 8.9	(m) 10.4	(n) 1.2	(o) 20	(p) 0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.0	(c) 37.4	(d) 20.9	(e) 36.3	(f) 20.8	(g) 35.2	(h) 20.9	(i) 24.0	(j) 29.8	(k) 23.5	(l) 29.2	(m) 23.0	(n) 28.7	(o) 4.2	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.5	18.7	25.5	22.3	18.4	25.4	22.0	18.1	25.2	75.4	30.0	73.4	29.2	71.9	28.3	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.9	8.1	7.4	-9.2	39.2	1.5	1.6	-10.4	40.3	-11.3	41.2	-12.1	42.1	-13.3	43.3		
			-10.1	24.8	1.5	0.5	-11.3	25.2	-12.2	25.5	-13.0	25.8	-14.2	26.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.0	8.1	11.0	15.5	19.5	23.4	27.5	28.2	28.3	24.6	19.4	13.1	8.7
	DBStd	8.32	4.52	5.39	5.02	4.29	3.84	2.43	2.14	2.28	3.04	4.80	4.83	4.49
	HDD10.0	261	90	49	14	1	0	0	0	0	0	3	24	80
	HDD18.3	1197	318	209	112	38	7	0	0	0	1	47	166	299
	CDD10.0	3536	31	76	185	285	414	524	564	568	439	293	117	39
	CDD18.3	1431	1	3	24	72	162	274	306	309	191	79	9	0
	CDH23.3	16809	21	87	283	783	1744	3344	3777	3852	1997	793	107	21
	CDH26.7	8160	2	18	69	276	778	1744	2020	2090	897	249	14	2
Wind	WSAvg	3.1	2.6	3.1	3.6	3.7	3.8	3.8	3.3	2.9	2.7	2.9	2.8	2.5
Precipitation	PrecAvg	543	25	30	41	45	67	69	50	64	89	55	36	19
	PrecMax	985	73	82	108	141	184	172	245	271	484	183	129	82
	PrecMin	347	0	0	0	2	7	0	3	0	1	1	0	1
	PrecStd	161	23	24	27	35	48	45	58	66	122	52	40	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.1	29.1	31.1	35.0	37.3	38.7	37.9	38.0	37.2	32.8	28.6	26.2
		MCWB	12.7	14.0	15.1	16.3	19.5	21.0	20.8	21.0	19.7	19.9	17.0	13.2
	2%	DB	22.8	26.1	28.7	32.3	35.0	37.2	37.2	37.4	34.1	31.2	26.3	22.7
		MCWB	12.4	13.7	14.9	16.7	20.0	20.9	21.0	21.2	20.7	19.7	16.6	13.3
	5%	DB	20.2	23.6	27.0	29.8	32.8	36.0	36.1	36.2	32.8	29.1	24.0	20.9
		MCWB	11.8	13.1	15.4	16.9	19.7	21.0	21.0	21.2	20.2	19.3	15.5	12.8
	10%	DB	17.7	21.3	24.1	27.6	31.2	33.9	34.7	34.9	32.0	27.7	22.4	17.8
		MCWB	10.8	13.0	14.8	16.8	19.6	21.0	21.1	21.1	20.2	18.6	15.1	12.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.8	18.5	19.7	21.4	23.8	24.7	24.4	24.2	24.0	23.3	20.9	18.3
		MCDB	19.8	21.4	23.6	26.5	29.7	31.3	30.6	29.9	29.3	27.6	24.6	20.1
	2%	WB	15.6	17.3	18.8	20.7	22.9	24.0	23.7	23.6	23.2	22.1	19.4	17.0
		MCDB	18.9	20.3	23.1	25.6	28.3	30.0	29.6	29.5	27.8	26.5	22.6	19.4
	5%	WB	13.7	16.2	18.0	20.1	22.3	23.4	23.2	23.1	22.7	21.4	18.1	15.1
		MCDB	18.0	20.0	22.0	24.9	27.7	29.3	28.9	29.2	27.3	26.0	21.0	18.1
	10%	WB	11.5	14.6	17.2	19.3	21.7	22.8	22.8	22.8	22.2	20.7	17.1	12.9
		MCDB	16.6	19.4	21.2	24.2	27.3	28.8	28.7	29.1	27.0	25.5	20.5	17.1
Mean Daily Temperature Range	5% DB	MDBR	15.1	15.2	14.4	14.3	12.5	11.9	11.7	12.0	12.0	13.7	14.9	14.6
		MCDBR	20.1	19.7	17.3	17.1	15.2	14.0	13.3	13.5	14.4	14.4	17.1	19.5
		MCWBR	11.0	10.5	8.0	7.7	5.6	3.8	3.8	3.7	4.9	6.1	9.1	10.8
	5% WB	MCDBR	14.5	13.4	11.8	11.9	11.3	11.0	11.1	11.1	10.0	10.5	12.3	13.3
		MCWBR	9.8	9.3	6.5	6.5	4.7	3.7	3.5	3.5	3.6	5.2	8.1	9.6
Clear-Sky Solar Irradiance	taub	0.286	0.292	0.314	0.348	0.370	0.381	0.393	0.378	0.365	0.330	0.305	0.288	
	taud	2.574	2.564	2.509	2.427	2.423	2.437	2.450	2.491	2.457	2.549	2.548	2.568	
	Ebn at Noon	958	980	972	943	918	902	890	903	906	923	925	937	
	Edh at Noon	82	91	103	116	117	115	113	108	107	91	84	79	
All-Sky Solar Radiation	RadAvg	3.32	4.06	5.16	6.16	6.42	7.08	6.93	6.43	5.31	4.36	3.47	3.00	
	RadStd	0.36	0.54	0.42	0.48	0.60	0.52	0.52	0.35	0.41	0.62	0.45	0.38	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page